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Neotectonic deformation in the western sector of tectonic escape in Anatolia: palaeomagnetic study of the Afyon region, central Turkey

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Abstract

Following final closure of the Neotethyan Ocean during the late Miocene, deformation in central Turkey has led to crustal thickening and uplift to produce the Anatolian Plateau followed by westward extrusion of terranes by strike–slip. Widespread volcanism has accompanied this latter (neotectonic) phase, and palaeomagnetic study of the volcanism shows a coherent record of differential block rotations, indicating that the Anatolian region is not a plate (or 'platelet') sensu stricto but is undergoing distributed internal deformation. To evaluate the scale of neotectonic rotations in the transition zone near the western limit of tectonic escape and the border of the extensional domain in central-west Turkey, we have studied the palaeomagnetism at 82 sites in volcanic suites distributed along a ~140-km lineament with north–south trend and ranging in age from 18 to 8 Ma. Comparable deflection of magnetic remanence from the present field direction is identified along the full length of the lineament. A mean clockwise rotation of $12.3 \pm 4.2°$ is determined for this western sector of the Anatolian strike–slip province. Since similar rotations are observed in the youngest and oldest units, this cumulative rotation occurred after the late Miocene. When interpreted together with results elsewhere in Anatolia, it is inferred that the rotation is later than crustal thickening and uplift of the Anatolian Plateau and entirely a facet of the tectonic escape. Inclinations are mostly $\sim 10°$ shallower than the predicted Miocene field and are considered to reflect the presence of a persistent inclination anomaly in the Mediterranean region. Larger rotations departing from the regional trend are also observed within the study region, but are confined to the vicinity of major faults, notably those bounding the Afyon-Akşehir Graben.

The pattern of neotectonic declinations across Anatolia identifies strong anticlockwise rotation in the east near the Arabian pincer with progressive reduction in the amount of rotation towards the west; it becomes zero or slightly clockwise at the western extremity of the accreted terrane collage. Rotations also appear to become generally younger towards the south. Crustal deformation has therefore been distributed, and the net effect of terrane extrusion to the west and south has been to expand the curvature of the Tauride Arc. The westward radial expansion of the extruded terranes is inferred to combine with backroll on the Hellenic Arc to produce the contemporary extensional province in western Turkey.
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Keywords: Neotectonics; Turkey; Anatolia; Afyon; Palaeomagnetism; Tectonic rotation; Tectonic escape

1. Introduction

Present-day Anatolia is part of the Alpine–Himalayan collisional belt and lies at the site of conver-

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gence between the Eurasian and Afro-Arabian plates. The accretionary zone comprises a collage of continental fragments which moved across the Palaeotethys Ocean from former sites at the northern margin of Gondwana with a new ocean (Neotethys) opening in their wake (Şengör and Yılmaz, 1981; Robertson et al., 1996). Closure of the Neotethys commenced during the Late Cretaceous, and subduction accompanied accretion of continental fragments along the Eurasian margin, mainly during Eocene times. By mid-Miocene times, the major tectonic elements of present-day Anatolia were in place following the collision of Arabia with this terrane collage along the Bitlis suture zone. Hence, the Cenozoic tectonic history is loosely divided into palaeotectonic and neotectonic phases with the former dominated by collisional tectonics and the latter comprising the post-collisional deformation that has resulted from crustal thickening and tectonic escape. This latter regime is driven by differential slippage between the African and Arabian sectors along the Dead Sea Fault Zone (DSFZ, see Fig. 1). A feature of this regime has been the concentration of strike–slip motions along intracontinental transforms including the dextral North Anatolian Transform Fault Zone (NAFZ) and the sinistral East Anatolian Transform Zone (EAFZ, Fig. 1). These fault zones form the main tectonic boundaries of a broad zone of distributed deformation within Anatolia.

The elastic deformation across this zone is taken up along two groups of second-order faults. The first set comprises NE–SW fractures that splay off mostly from the NAFZ into the Anatolian collage; the largest examples are the Kırıkkale-Erbaa and Almus fault zones shown in Fig. 1. Members of this system can display both sinistral and dextral strike–slip move-

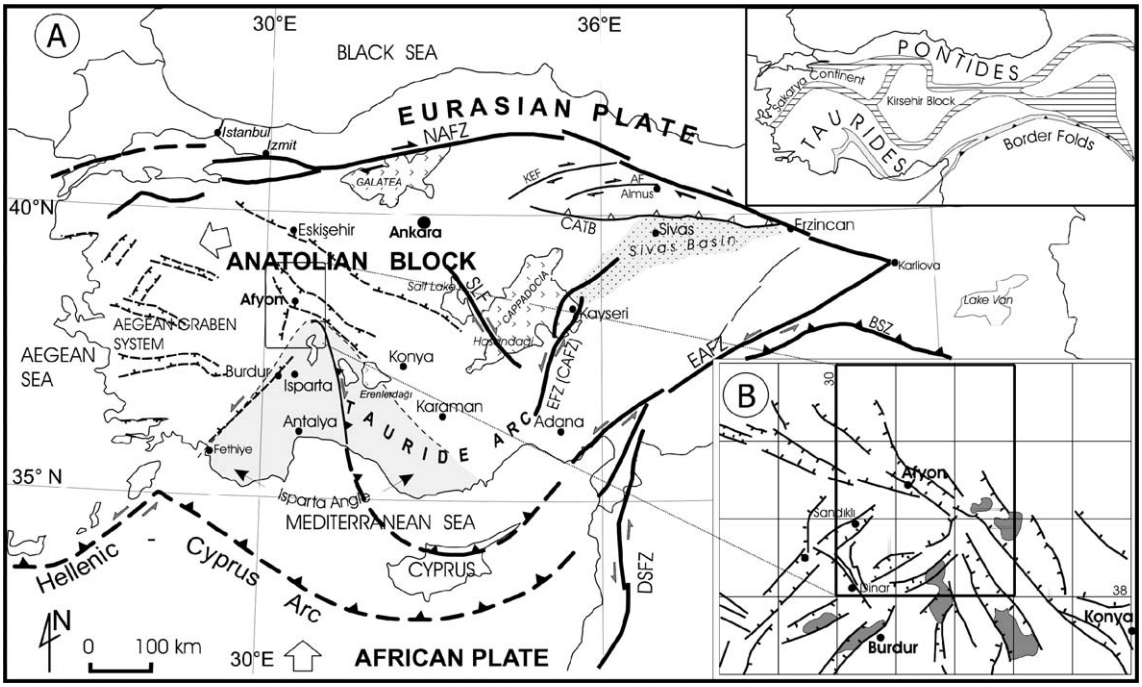


Fig. 1. The tectonic divisions and distribution of major lineaments in Turkey and adjoining regions. The large open arrows show current relative motions in western Turkey and the eastern Mediterranean, and the smaller half arrows are directions of movement on major strike–slip faults. The abbreviations are SLF, Salt Lake Fault Zone; KEF, Kırıkkale-Erbaa Fault Zone; AF, Almus Fault Zone; EFZ (CAFZ), Ecemiş Fault Zone (Central Anatolian Fault Zone); CATB, Central Anatolian Thrust Belt; and DSFZ, Dead Sea Fault Zone. The region of the present study in the vicinity of the City of Afyon and close to the apex of the Isparta Angle is amplified in the inset (B) with the major faults delineating the Afyon-Aksehir Graben. The shaded area comprises the Isparta Angle and was formed mainly during the Palaeotectonic era by interference of verging allochthonous units during final stages of Tethyian convergence.

ments, plus normal or reverse-slip components, where there are variations from this general trend (Kocyiğit and Beyhan, 1998), but left-lateral strike–slip is dominant in the east of Anatolia. A second set of faults has NW–SE trends and oblique-slip motions, usually with a dominant right-lateral motion. Members of this set divide the western half of central Anatolia into horsts and grabens bounded by active faults, and they are concentrated within a transitional tectonic zone between the strike–slip regime of eastern Anatolia and the extensional regime of the Aegean and western Anatolia (Yılmaz et al., 1987 and Fig. 1).

Several methods can be applied to quantify neotectonic crustal deformations and apply to three contrasting time scales: in the short term ($10–10^2$ years), kinematics may be resolved from GPS and ground surveying. On archaeological to late geological time scales ($\sim 10^3–10^5$ years), deformation can be evaluated by geomorphic study of such features as stream offsets and changing elevations of erosional and depositional surfaces. In the longer term ($>10^5$ years), recovery of palaeomagnetic directions is usually required to quantify crustal deformation and has been effectively applied over much of Anatolia due to the widespread preservation of young volcanic rocks that tend to be good recorders of the palaeofield direction. This method aims to resolve cumulative kinematics by comparing palaeomagnetic directions with reference directions of the same age from the adjoining major continental plates. Contrasts in magnetic declination are the signature of cumulative tectonic rotation since the age of magnetisation. In principle, contrasts in magnetic inclination (following adjustment for any tectonic tilt) are the signature of relative latitudinal movement. However, in practice, these latter differences are more difficult to apply on time scales covered by the neotectonic history because geomagnetic inclination has, in the majority of cases, been found to be too shallow along the Alpine–Himalayan orogen to be accommodated by actual latitudinal movements (Beck et al., 2001; Piper et al., 2002); the tectonic significance of differences in magnetic inclination cannot therefore be precisely quantified. Combining geological observation with distributed palaeomagnetic sampling may resolve the size and integrity of the fault blocks involved in the rotation.

Palaeomagnetic studies in the eastern part of Anatolia have demonstrated a regional pattern of distributed block rotations (Gürsoy et al., 1997, 1998, 1999; Piper et al., 1996, 1997, 2001; Platzman et al., 1994, 1998; Tatar et al., 1996, 2000, 2002). These appear to be a late facet of the neotectonic history and have followed crustal thickening in late Pliocene and Quaternary times (Gürsoy et al., 1997; Tatar et al., 2002; Piper et al., 2002). The differential character of this deformation is illustrated by a decrease in the amount of anticlockwise rotation southwestwards towards the horst and graben province of western Turkey (Fig. 1). West of the longitude of Ankara rotations are variable, but declinations recovered to date are consistently clockwise from the Galatean region north of Ankara to the Isparta Angle close to Afyon in the south (Gürsoy et al., 1999; Tatar et al., 2002). The Isparta Angle (Fig. 1) is a tectonic bend formed primarily during the palaeotectonic history by interference of verging allochthonous units during culminating stages of Tethyian convergence. To the east of this longitude, rotations become insignificant and then progressively more anticlockwise towards the Arabian pincer compatible with the westward and southward extrusion of crust from this zone of indentation during the last few million years.

The cumulative rotations determined from palaeomagnetism are unlikely to have resulted from a uniform stress regime of long continuity because there appears to have been a progressive clockwise rotation of the regional stress field across Anatolia during Pliocene and Quaternary times. This is evident both in the Ankara area, where regional compression moved from NW–SE before late Pliocene times to N–S during Quaternary times (Kocyiğit, 1991), and in western Anatolia, where regional extension moved from N–S to NE–SW over broadly the same period (Glover and Robertson, 1998). Results from the Global Positioning System (GPS) show that recent surface motions in Anatolia can be approximated by a clockwise rotation fitted by an Eulerian pole in Sinai (Barka and Reilinger, 1997; McClusky et al., 2000). This would be compatible with general westward extrusion of the crust by tectonic escape and is expressed by dominant strike–slip within a broad sector of central Anatolia extending westwards from the triple junction at Karlıova where the NAFZ and EAFZ meet (Fig. 1). Currently, this region is moving

westwards relative to Eurasia at a rate of ~15 mm/year and western Anatolia (i.e., west of the Isparta Angle) is moving southwest at ~30 mm/year. The near-static behaviour of the Isparta Angle region (Barka et al., 1999) separates an eastern sector dominated by a regime of Pliocene and later compression and strike–slip, from a dominantly extensional regime within the eastern part of the Aegean province (Fig. 1). The extensional regime in western Turkey is probably driven predominantly by rollback of the Hellenic Arc (LePichon and Angelier, 1979) that has been progressively expanding its curvature during the last 5 Ma (Kissel et al., 1989). Neotectonic deformation in the Isparta Angle therefore appears to be currently controlled by the interaction of back arc processes associated with outward motion of the Hellenic Arc in the west and the extrusion of Anatolian crust to the east.

No systematic palaeomagnetic studies have yet investigated crustal deformation in this zone of transition between the strike–slip and extensional regimes. As a contribution to resolving neotectonic deformation in this region, the present study has been undertaken over an extensive volcanic region–extending north and south of the city of Afyon. This region lies astride the western part of the strike–slip domain and abuts against the eastern margin of the Isparta Angle (Fig. 1). The palaeomagnetic analysis has aimed to address the degree to which deformation across this zone is distributed and determine how the long-term picture of deformation as determined from palaeomagnetism compares with the short-term record resolved by GPS.

2. Tectonic and regional geological framework

The Neogene evolution of Anatolia has been discussed in the context of four tectonic models. They are not mutually exclusive, and it is likely that several driving forces, operating to differing degrees and over contrasting time periods, have produced the prevailing tectonic framework.

(i) The *tectonic escape model* invoked by many authors since McKenzie (1972) considers that west to southwestward expulsion of central and eastern Anatolia followed collision between the Afro-Arabian and European plates at ~12 Ma. The extensional horst–graben system in the region to the west is a possible consequence of the expansion of this extruding terrane assemblage.

(ii) The *back-arc spreading model* proposes that south-southwestward migration of the Hellenic Trench is responsible for the Aegean extensional domain. Since estimates for the onset of Hellenic subduction range up to 60 Ma (LePichon and Angelier, 1979; Meulenkamp et al., 1988), this model would allow for long-lived extension in a regime of considerable antiquity.

(iii) The *orogenic collapse model* (Dewey, 1988; Seyitoğlu and Scott, 1996) proposes that the neotectonic regime is the consequence of lateral spreading of continental crust. This model predicts that the onset of horst–graben formation in western Anatolia occurred during the late Oligocene–early Miocene following cessation of the Palaeocene collision along the Izmir–Ankara–Erzincan Neotethyan suture zone (IAESZ, the location of this suture is shown on our summary, Fig. 8) and the associated shortening and overthickening of Aegean–Anatolian crust. This model further predicts that graben formation has been continuous since the late Oligocene–early Miocene.

(iv) The *two-stage graben model* postulates that back-arc spreading, orogenic collapse, tectonic escape, and rollback processes have all operated successively, or in tandem, to form the graben–horst systems (Koçyiğit et al., 1999, 2000). This model is based on studies of rift infills in western Anatolia that recognise deposition in extensional regimes during early–middle Miocene and Pliocene–Recent times with these two phases of sedimentation interrupted by late Miocene–early Pliocene compression. The earlier phase of graben extension is reflected by deposition that appears to be temporally related to orogenic collapse along the IAESZ (Dewey, 1988; Seyitoğlu and Scott, 1996). This early extensional phase is separated from the later extensional phase by a short interval of ENE–WSW-directed compression during late Miocene times (Koç yiğit et al., 1999). The latter phase was contemporaneous with collision along the Bitlis suture zone between the Afro-Arabian plate and Anatolian collage (Fig. 1) and was associated with the latest convergence, thickening, and uplift of the Anatolian region (Dewey et al., 1986). By early Pliocene times, seafloor spreading had commenced along the axis of the

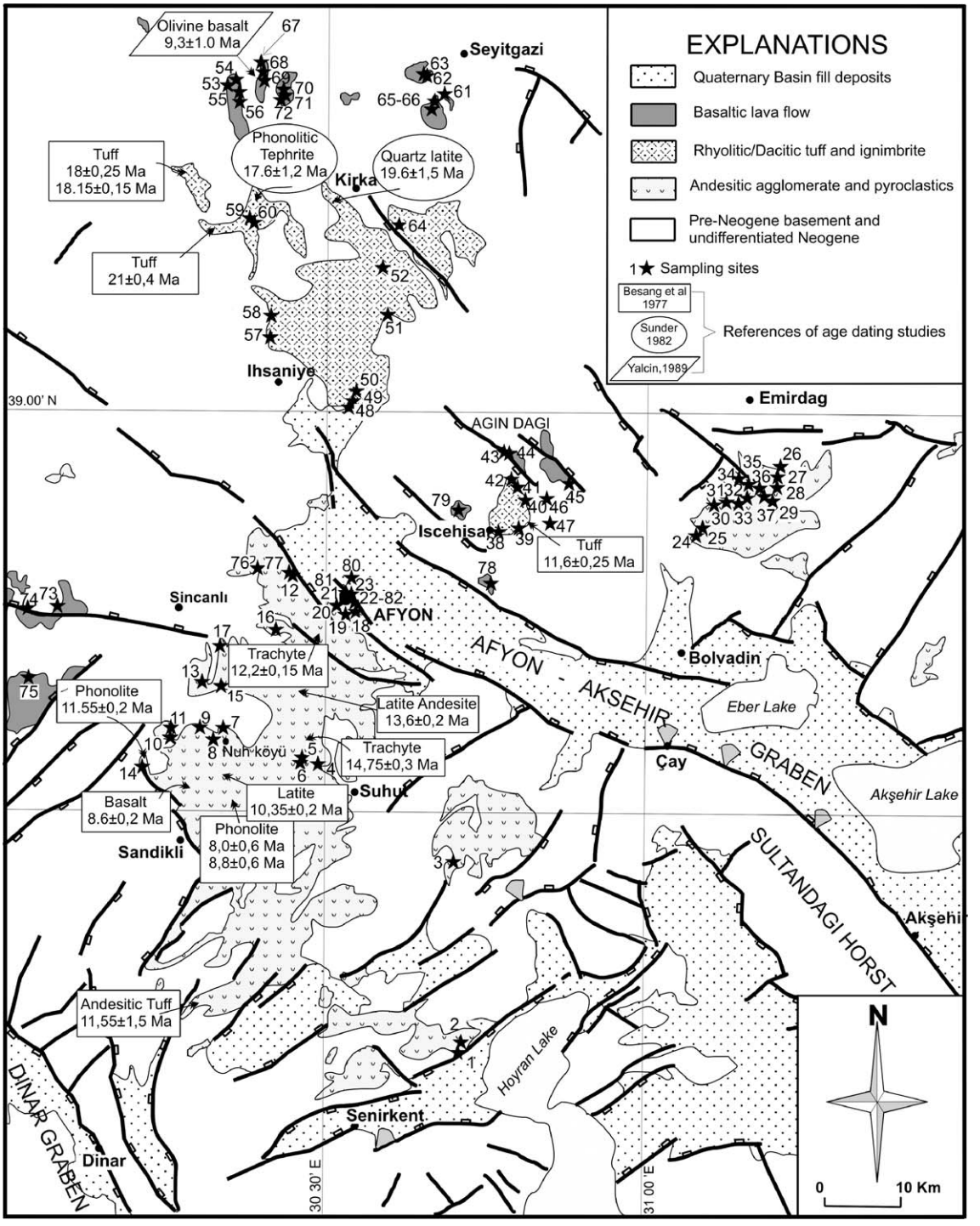


Fig. 2. Outline geological map of Afyon region (see boxed area in Fig. 1B) showing the distribution of major volcanic outcrops and graben sedimentary infill. The stars and numbers show the palaeomagnetic sampling sites of this study. The boxes and ovals show locations of age determinations on the volcanics.

Red Sea to produce the differential northward movement of Afro-Arabia (Hempton, 1987). This initiated the westward extrusion of the Anatolian terrane collage and has been contemporaneous with the last phase of extension in western Anatolia.

The consumption of the Neotethyan Ocean along the IAESZ caused the major environmental change recorded throughout Anatolia, which occurred at about 11.8 Ma during latest Serravallian–earliest Tortonian times. Shallow-water Serravallian sediments are the last marine record and were succeeded by terrestrial sediments as central Anatolia became an upland region characterised by horsts and fault-bounded depressions (Ovas). Quasi-continuous volcanism has recorded the waning stages of subduction and subsequent fault movements, and it falls broadly into three (western, central, and eastern) provinces (Yılmaz, 1990). The Central Anatolian Volcanic Province (CAVP) embraces the volcanic rocks of this study and lasted from Eocene to Pliocene times. Volcanism in the CAVP was mostly calc-alkaline in character until the late Miocene (Keller, 1983; Aydar, 1998). Since then, it has included both alkaline and calc-alkaline magmas with the changing balance of lava compositions probably related to a transition in tectonic regimes (Yılmaz, 1990). In the Afyon region, volcanic activity decreases in age from ~ 20 Ma in the north to ~ 4 Ma in the south and is sited along half grabens bounded by left-lateral NE–SW and right-lateral NW–SE fault sets bordering the Isparta Angle (Savaşcın et al., 1995).

Volcanic outcrop in the Afyon region comprises a number of dispersed fields of mixed acid and basic composition (Boray et al., 1985; Harut, 1995; Ronner, 1962). Most activity is focussed along a ~ 140-km-long south to north lineament running through the city of Afyon and extending from the northern flanks of the Dinar Graben bordering the Isparta Angle in the south to the vicinity of Kırka in the north (Fig. 2). Although the deep origin of this lineament is presently unclear, a close proximity to the boundary between the strike–slip and extensional provinces is apparent. The volcanics have both alkaline and calc-alkaline affinities (Keller and Villari, 1972; Çevikbaş et al., 1988), and volcanic domes in the vicinity of Afyon are intercalated with tuffs, agglomerates, volcano-sedimentary deposits, and palaeosols of middle to late Miocene age (Öztürk, 1989). South of Afyon, the volcanics include leucite, phonolite, and lamproite suites cutting an older trachyte–rhyolite sequence. North of Afyon, thick tuff and ignimbrite deposits are interleaved with rhyolitic lavas and terminate in latites, trachytes, and phonolitic tephrites (Keller and Villari, 1972). Besang et al. (1977) dated the trachytes and phonolitic tephrites in this area at 21–17 Ma.

The stratigraphic and radiometric age constraints identify intermittent volcanic activity between middle and late Miocene times (Besang et al., 1977; Sunder, 1982; Yalçın, 1989). K–Ar determinations range from 14.75 ± 0.3 to 8.0 ± 0.6 Ma (Besang et al., 1977; Becker-Platen et al., 1977). The youngest (8.0 Ma) age is from volcanics within the Akin Formation. Early middle Miocene mammalian-bearing horizons correlate with the lower levels of this formation, and the highest levels are dated as late Miocene from similar faunas (Sickenberg et al., 1975).

3. Field and laboratory methods

Each lava unit (Table 1) was sampled, typically as seven independent cores, drilled over several metres of outcrop using a portable motor and oriented by both sun and magnetic compasses. Lavas are typically high-quality recorders of the palaeomagnetic field direction at the time of eruption. Rapid cooling and weak fabrics will usually mean that they record a geomagnetic field direction that is instantaneous in geological terms and therefore a component of the secular variation; sampling of a number of flows is therefore always required to yield a mean direction of palaeomagnetic, rather than virtual geomagnetic, significance. The attitude of flow structures and pseudo-bedding in lavas and of layering in the pyroclastic units was routinely measured. However, indications of bedding were not recognisable at some massive outcrops; elsewhere, dips contrasting significantly with the palaeohorizontal (>10°) were evident at only 20 sites (Table 1). The application of tilt adjustment to the palaeomagnetic data is unclear because the volcanic units have often flowed down primary slopes, and when this is the case, most or all of the measured tilt is primary, and tilt adjustment is not relevant. In this study, full tilt adjustments were

Table 1
Palaeomagnetic sampling sites in volcanic rocks of the Afyon region, Central Anatolia

Site	Rock type	Age (with footnote reference)	Location	N	Mo	Tilt
1	ignimbrite	L. Miocene–Pliocene (4, 5)	Hoyran Lake	7	9.2–12.9	15/169
2	ignimbrite	L. Miocene–Pliocene (4, 5)	Hoyran Lake	7	99.5–228.3	H
3	basalt	L. Miocene–Pliocene (4, 5)	SE Şuhut	8	74.9–124.8	8/270
4	basalt	L. Miocene–Pliocene (1, 2, 9)	NW Şuhut	9	11.9–34.8	16/010
5	basalt	L. Miocene–Pliocene (1, 2, 9)	NW Şuhut	5	64.5–100.8	H
6	basalt	L. Miocene–Pliocene (1, 2, 9)	NW Şuhut	6	182.4–1398.8	19/070
7	basalt	L. Miocene–Pliocene (1, 2, 9)	Nuhköy	7	25.2–58.9	21/066
8	basalt	L. Miocene–Pliocene (1, 2, 9)	Nuhköy	6	35.7–56.7	10/388
9	basalt	L. Miocene–Pliocene (1, 2, 9)	Nuhköy	7	64.9–137.2	H
10	basalt	L. Miocene–Pliocene (1, 2, 9)	Ciğiltepe	7	30.8–65.5	H
11	basalt	L. Miocene–Pliocene (1, 2, 9)	Ciğiltepe	7	44.3–56.5	H
12	andesite	M.–L. Miocene (2, 9)	W. Afyon	6	74.4–126.3	H
13	andesite	Pliocene (7)	Nuh Village	7	14.0–24.3	H
14	basalt	L. Miocene–Pliocene (1, 2)	Karadirek	7	204.1–338.5	H
15	basalt	L. Miocene–Pliocene (1, 2)	Taşoluk	7	19.8–96.6	H
16	basalt	L. Miocene–Pliocene (1, 2)	Kayadibi	7	9.2–16.6	H
17	basalt	L. Miocene–Pliocene (1, 2)	Sinirköy	7	29.5–60.9	H
18	andesite	M.–L. Miocene (2, 9)	Hıdırlık Tepe	7	94.0–230.4	H
19	andesite	M.–L. Miocene (2, 9)	Hıdırlık Tepe	6	34.9–124.2	19/190
20	andesite	M.–L. Miocene (2, 9)	Afyon Castle	7	51.8–71.7	–
21	andesite	M.–L. Miocene (2, 9)	Afyon Castle	7	17.5–20.3	–
22	andesite	M.–L. Miocene (2, 9)	Afyon	7	46.7–62.5	–
23	andesite	M.–L. Miocene (2, 9)	Afyon	7	28.8–69.1	–
24	andesite	M.–L. Miocene (9)	E. Afyon, Bolvadin	7	262.1–823.5	H
25	andesite	M.–L. Miocene (9)	E. Afyon, Bolvadin	7	0.31–9.34	H
26	andesite	M.–L. Miocene (9)	S. Emirdağ	7	79.1–98.2	3/180
27	andesite	M.–L. Miocene (9)	S. Emirdağ	7	79.9–205.5	H
28	andesite	Pliocene (9)	S. Emirdağ	7	39.1–113.2	H
29	andesite	Pliocene (9)	S. Emirdağ	7	50.7–169.2	H
30	andesite	M.–L. Miocene (9)	E. Afyon, Kemerkaya	7	40.9–51.9	H
31	andesite	M.–L. Miocene (9)	E. Afyon, Kemerkaya	7	83.3–123.2	H
32	andesite	M.–L. Miocene (9)	E. Afyon, Kemerkaya	7	41.0–203.9	21/297
33	andesite	M.–L. Miocene (9)	E. Afyon, Kemerkaya	7	16.0–26.7	H
34	andesite	M.–L. Miocene (9)	Çatallı village	7	167.7–283.5	H
35	basalt	Pliocene (9)	Çatallı village	7	71.6–101.5	H
36	andesite	Pliocene (9)	Guzleyayla	7	35.5–81.7	H
37	andesite	M.–L. Miocene (9)	Guzleyayla	7	53.3–123.5	H
38	basalt	Mio-Pliocene (9)	İşcehisar	7	29.1–88.9	H
39	basalt	Mio-Pliocene (9)	İşcehisar	7	90.8–164.0	H
40	ignimbrite	Mio-Pliocene (9)	Karakaya	7	*	4/180
41	ignimbrite	Mio-Pliocene (9)	İşcehisar	7	*	25/109
42	ignimbrite	Mio-Pliocene (9)	İşcehisar	7	*	24/85
43	andesite	Mio-Pliocene (9)	NE Afyon, İşcehisar	7	37.1–57.2	H
44	ignimbrite	Mio-Pliocene (9)	İşcehisar	7	37.5–50.4	H
45	ignimbrite	Mio-Pliocene (9)	İşcehisar	7	0.93–2.10	10/030
46	ignimbrite	Mio-Pliocene (9)	İşcehisar	7	*	11/165
47	andesite	Mio-Pliocene (9)	İşcehisar	7	138.1–426.8	H
48	ignimbrite	Miocene (9)	Kırka road	7	17.6–27.2	10/180
49	acid tuff	Miocene (9)	Kırka road	7	2.8–4.1	H
50	ignimbrite	Miocene (9)	Kırka road	7	20.6–29.2	18/225
51	dacite	Miocene (9)	Kırka road	7	3.2–4.7	H
52	ignimbrite	Miocene (9)	Kırka road	7	7.2–10.8	26/246

(continued on next page)

Table 1 (*continued*)

Site	Rock type	Age (with footnote reference)	Location	N	Mo	Tilt
53	andesite	Miocene (6)	Türkmendağı	7	5.3–24.6	H
54	andesite	Miocene (6)	Türkmendağı	7	76.2–1115.9	H
55	andesite	Miocene (6)	Türkmendağı	7	21.2–42.9	H
56	andesite	Miocene (6)	Türkmendağı	7	55.5–98.6	H
57	ignimbrite	Mio-Pliocene (9)	N. Afyon, İhsaniye	7	19.4–23.7	H
58	ignimbrite	Mio-Pliocene (9)	N. Afyon, İhsaniye	7	44.2–57.1	11/040
59	ignimbrite	Mio-Pliocene (9)	Kırka road	7	8.6–11.6	11/210
60	ignimbrite	Mio-Pliocene (9)	Kırka road	7	2.7–4.6	11/212
61	basalt	Miocene (6)	SW Seyitgazi	7	28.8–219.3	10/150
62	basalt	Miocene (6)	SW Seyitgazi	7	29.7–171.6	10/152
63	basalt	Miocene (6)	SW Seyitgazi	7	24.4–39.1	10/151
64	ignimbrite	Mio-Pliocene (9)	N. Afyon, SE Kırka	7	34.2–42.4	22/202
65	basalt	Miocene (6)	SW Seyitgazi	7	74.2–1734.0	10/056
66	basalt	Miocene (6)	SW Seyitgazi	7	15.3–28.5	10/054
67	andesite	Miocene (6)	Türkmendağı	7	2.1–3.1	H
68	andesite	Miocene (6)	Türkmendağı	7	9.5–25.6	H
69	andesite	Miocene (6)	Türkmendağı	7	60.7–93.7	H
70	andesite	Miocene (6)	Türkmendağı	7	16.7–47.8	26/285
71	basalt	Miocene (6)	Türkmendağı	7	5.5–15.0	45/054
72	basalt	Miocene (6)	Türkmendağı	7	22.1–237.9	H
73	basalt	Mio-Pliocene (9)	W. Sincanlı	7	15.7–56.7	29/199
74	basalt	Mio-Pliocene (9)	W. Sincanlı	7	26.8–54.2	H
75	basalt	Mio-Pliocene (9)	W. Sincanlı	7	23.7–42.7	H
76	andesite	Mio-Pliocene (9)	Köprülü village	7	87.9–129.3	H
77	andesite	Mio-Pliocene (9)	Afyon centre	7	80.1–2306.7	20/203
78	basalt	Mio-Pliocene (9)	Gebeciler	7	75.0–221.8	–
79	basalt	Mio-Pliocene (9)	W. İşcehisar	7	12.8–30.7	–
80	andesite	Mio-Pliocene (9)	Ihsaniye road	7	119.0–160.8	11/266
81	andesite	Mio-Pliocene (9)	Afyon centre	7	44.4–65.3	–
82	andesite	Mio-Pliocene (9)	Afyon centre	7	17.5–42.0	–

H = horizontal. References to ages deduced from stratigraphic, morphological, and K–Ar evidence are 1, Ronner (1962); 2, Besang et al. (1977); 3, Sunder (1982); 4, Koçyiğit (1984); 5, Boray et al. (1985); 6, Yalçın (1989); 7, Harut (1995); 8, Aydar (1998); 9, author's field observations. NRM intensities of magnetisation are in units of $\times 10^{-4}$ A m^2/kg. *Sites not demagnetised due to very weak total NRMs.

evaluated, but only where they brought the magnetic inclination closer to the predicted field inclination in this area ($I = 58.5°$) were the tilt-adjusted directions preferred.

In the laboratory cores were sliced into 2.3-cm length cylinders and Natural Remanent Magnetisations (NRMs) measured by "Minispin" magnetometers. NRM intensities range by two orders of magnitude (Table 1), but often vary as much within sites as they do between sites. The only apparent correlation with petrology is that tuffaceous units are more weakly magnetised than lavas. About half of the cores were typically subjected to progressive alternating field (AF) demagnetisation in steps of 5 to 30 mT and then in steps of 10 mT, until directional behaviour or intensity change ceased to be systematic. The remainder of the collection was subjected to thermal demagnetisation in MM2 D demagnetisers in steps of 50 or 100 to 500 °C and then in steps of 20 °C to the Curie points of the magnetic carriers. The demagnetisation procedure was conducted within a neutralised-field space provided by large sets of Rubens coils to minimise secondary remanence acquisition. Orthogonal projections of the magnetisation vector during the progressive cleaning were produced for each sample to isolate the components comprising the NRM and calculate their equivalent directions by principal component analysis. No systematic differences were recognised in these collections between the Characteristic Remanent Magnetisations (ChRMs) defined by the two treatment procedures and common site populations resolved by both methods have been combined

to yield the site mean results listed in Table 3. Lowest blocking temperature/coercivity components are laboratory or drilling-related acquisitions, or are longer term viscous remanence in the present field; they are removed in the first few steps of treatment (see Figs. 4 and 5) and are not included in this table.

4. Rock magnetic results

The ferromagnets in basaltic and andesitic volcanics are typically titanomagnetites with compositions close to the middle of the magnetite–ulvöspinel solution series. Although such grains would have low Curie points of ~250 °C, they usually undergo sub-solidus exsolution during magmatic, deuteric, and/or hydrothermal stages to produce ilmenite lamellae hosted by Ti-poor magnetite. Subsequent oxidation by reaction with high-fugacity fluids may incipiently oxidise these minerals to hematite and pseudobrookite. The magnetic consequence is to raise the Curie point close to that of pure magnetite and subdivide the magnetic grain size and increase single-domain (SD) and pseudo-single-domain (PSD) structures. The latter effect enhances magnetic stability, although since these reactions are usually concentrated within late stages of cooling, the recorded palaeofield direction is unlikely to differ from the time of eruption.

Rock magnetic tests are used to identify the ferromagnetic carriers and their dominant domain states. Results for samples from 20 lava sites for which both Curie point and hysteresis data are available are summarised in Table 2. Strong field thermomagnetic analysis measures the variation in saturation magnetisation (Js) with temperature to determine the Curie temperature and, hence, titanomagnetite composition. Saturation magnetisation was measured from room temperature to 700 °C using a computer-controlled horizontal Curie balance. Curie temperatures were estimated from the thermomagnetic curve using the method of intersecting tangents, and the change in saturation magnetisation at 100 °C following heating and cooling (the ratio RM in Table 2) is a measure of alteration produced by heating in air.

In the lavas, a clear thermomagnetic signature of titanomagnetite close to pure magnetite in composition (Curie point ~585 °C) is usually observed. A few sites have dual Curie points in the titanomagnetite range; the lower one may be due to unmixing of ulvöspinel, although the Curie temperature is sometimes low enough (19 and 20 in Table 2) to be attributable to a nonstochiometric titanomaghemite (cation-deficient titanomagnetite). Saturation remanence is always reduced in the lavas by heating in air due to partial oxidation of the magnetite to hematite. Thermomagnetic behaviours in the pyroclastic rocks are more complex (Fig. 3) due either to processes occurring during air fall or to reactions following emplacement. In these units, hematite is a significant or sole contributor to the thermomagnetic spectra. It may also be present in small amounts in the lavas, but since its saturation remanence is much

Table 2
Thermomagnetic and hysteresis results, volcanic rocks of rocks of the Afyon region, Central Anatolia

Site number	Tc1	Tc2	RM	M_s	M_{rs}	H_c	M_{rs}/M_s	X_{MD}
4	–	570	0.91	1.30	0.12	11.44	0.092	0.85
5	–	575	0.74	0.79	0.11	14.11	0.141	0.75
6	–	590	0.57	1.46	0.15	8.99	0.105	0.82
7	430	590	0.52	0.33	0.06	17.64	0.170	0.69
8	400	570	0.50	1.14	0.15	14.00	0.132	0.78
9	–	575	0.67	1.37	0.20	13.72	H	–
10	–	575	0.67	1.29	0.20	16.64	0.155	0.72
11	–	580	0.56	1.54	0.24	15.98	0.157	0.71
12	–	590	0.63	1.28	0.22	16.77	0.168	0.69
13	–	575	0.93	0.92	0.16	21.86	0.177	0.67
14	–	590	0.37	1.01	0.14	11.04	0.135	0.76
15	–	–	–	1.76	0.22	11.44	0.127	0.78
16	–	585	0.40	0.96	0.09	9.50	0.091	0.85
17	500	560	0.79	0.70	0.21	31.48	H	–
18	–	580	0.48	0.77	0.13	17.18	0.173	0.68
19	380	560	0.45	1.00	0.13	14.07	0.134	0.76
20	385	590	0.33	2.48	0.39	12.09	0.159	0.71
21	–	610	0.07	0.10	3.30	34.52	H	–
22	410	585	0.79	3.57	0.69	18.65	0.194	0.64
23	420	580	0.33	1.53	0.19	12.53	0.129	0.77

Tc1 and Tc2 are successive Curie points recognised by thermomagnetic study. RM is the ratio at 100 °C of the saturation magnetisation after cooling to the magnetisation before cooling. M_s, M_{rs} and H_c are the magnetisation in a saturating field, the residual saturation remaining when this field is removed, and the coercivity, respectively. Units of M_s and M_{rs} are A m^2/kg $\times 10^{-2}$ and H_c is in millitesla. Values of M_s and M_{rs}/M_s are not given if the sample failed to saturate in a 1-T field; this behaviour is probably due to a significant presence of hematite (=H). X_{MD} is an estimate of the ratio of MD to SD magnetite grains in the sample.

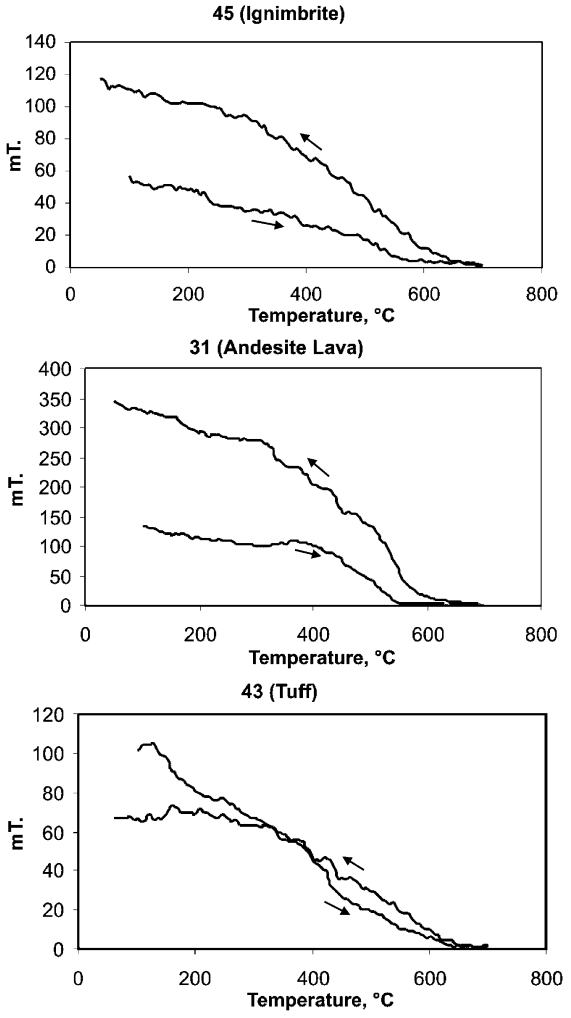


Fig. 3. Examples of thermomagnetic curves (saturation magnetisation, M_s, against temperature in °C) from volcanics of this study. The cooling curves are shorter and terminate at 100 °C. Lava at Site 31 is a typical low-Ti magnetite Curie point with a paramagnetic "tail," whereas the tuff of Site 43 identifies only titanohematite. The ignimbrite of Site 45 shows that both magnetite and hematite are present with the former partially oxidising to the latter during cooling; the hematite probably results from partial oxidation of the magnetite during post-emplacement consolidation and does not contribute significantly to the palaeomagnetic signature.

lower than magnetite, it is only detected by hysteresis study (Table 2).

Hysteresis analysis yields information relevant to understanding mineralogy and domain states of the ferromagnetic minerals: titanomagnetites and titanohematites saturate in fields of 300 mT or less, whilst titanohematites require much larger fields to achieve saturation. Using a Molspin vibrating sample magnetometer (VSM), hysteresis parameters comprising saturation magnetisation, M_s, saturation remanence, M_{rs}, and coercive force, H_c, were determined. With a maximum field of 1 T, it is not possible to fully saturate titanohematites with the VSM and M_s cannot be determined when this is a significant part of the ferromagnetic fraction. Parameters M_s and M_{rs} are dependent on the concentration and type of magnetic minerals present: M_s is independent of grain size, but M_{rs} is smaller for multidomain (MD) than for single-domain (SD) grain properties. Hence, the ratio of M_{rs}/M_s is a useful indicator of domain states: if $M_{rs}/M_s < 0.1$, the sample is dominated by MD grains; if $M_{rs}/M_s > 0.1$, small but significant fractions of SD grains are present within a dominant MD assemblage. Provided that these mixed domain sizes are present only within magnetite (as indicated by M_{rs}/M_s values of 0.02–0.5), the equation $X_{MD}=(0.5-(M_{rs}/M_s))/0.48$ yields an estimate of the fraction of MD grains present (Thomas, 1992).

Hysteresis results for lava sites 4–20 summarised in Table 2 indicate mixed SD and MD assemblages with the MD typically dominant, although there are always significant fractions of SD grains able to account for the stable remanence found in most samples. This general pattern of high MD/SD grain ratios has been widely recognised in the Miocene and younger lavas from Anatolia (Gürsoy et al., 1998; Tatar et al., 2002).

5. Palaeomagnetic results

Stepwise alternating field (AF) and thermal demagnetisation of lava and tuff samples from 82 sites (Table 1 and Fig. 2) yields vector plots dominated by single ChRMs of probable primary origin acquired during initial cooling. Examples of demagnetisation results are shown in Figs. 4 and 5 and are representative of the high quality of component definition at most sites. AF demagnetisation (Fig. 4) removes a variable viscous magnetisation in the first one to three steps to recover single components that decay to the origin. Thermal demagnetisation (Fig. 5) is subject to slightly more noise, possibly because incipient oxidation is induced in the furnaces, but better discriminates

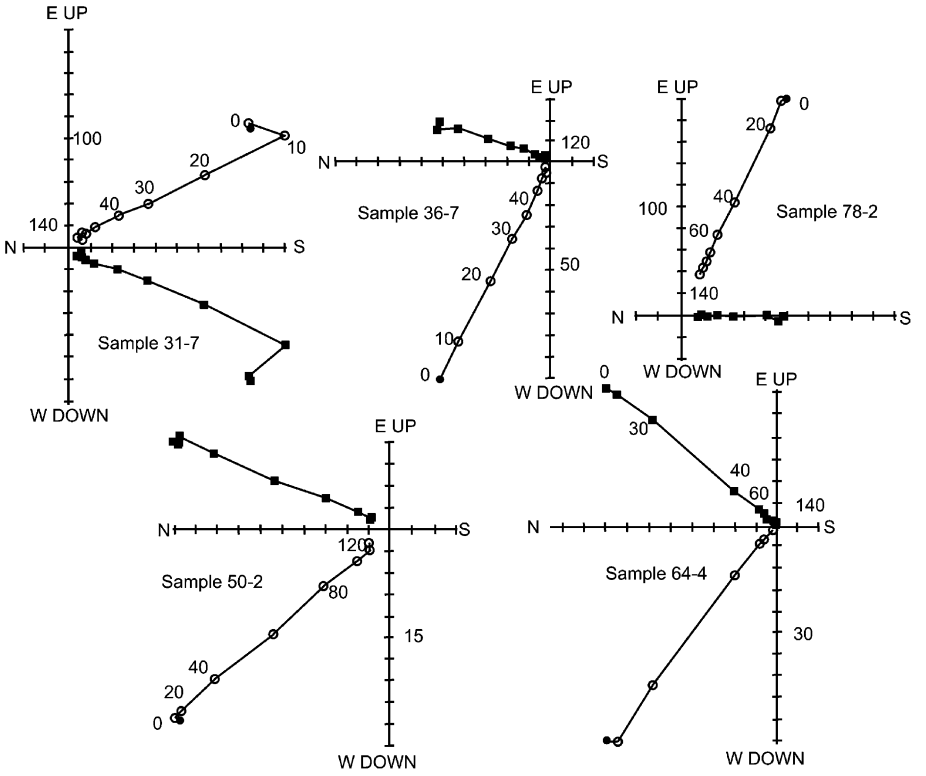


Fig. 4. Examples of typical AF demagnetisation results from samples of this study shown as orthogonal projections of the remanence vector onto the horizontal (closed squares) and vertical (open circles) planes. The demagnetisation steps are in units of millitesla (mT), and the intensity units are $\times 10^{-5}$ A m^2/kg.

lower blocking temperature components. Samples 25-5 and 53-1 in Fig. 5 show examples of components of normal polarity, presumably viscous acquisitions in the present field, removed to recover ChRMs of reversed polarity. This component definition is preserved at site level with consistent groupings and α_{95} usually less than 10° (Tables 3).

The palaeomagnetic directions may be compared with the present mean regional dipole field direction ($D/I = 0/+58.5°$). Directions within 40° of this mean are determined from 59 of the volcanic units. The remaining sites are regarded as anomalous due to a tectonic or geomagnetic cause (see below). Group mean directions derived from the dipolar groupings are summarised for each of five volcanic fields in Table 4 (see Figs. 2 and 7) and are rotated clockwise from the present-day field. During this latter part of Neogene times, the bordering major plates of Eurasia and Afro-Arabia have also rotated slightly clockwise but by lesser amounts (Besse and Courtillot, 1991). This is reflected in the predicted palaeofield directions at the study location computed from the mean pole positions of these major plates at 10 Ma (Table 5), and the contrasts in time-averaged declination with estimates of the regional field declination are a record of tectonic rotation. The majority of sites are clearly of normal or reversed magnetisation with inclinations somewhat shallower than the mean regional Miocene field and declinations rotated by small amounts clockwise (Table 3 and Fig. 6).

Some sites deviate from the dipolar field to an extent that identifies either magnetisation during a polarity transition or larger local tectonic rotation about an inclined axis. Since polarity transitions are rapid events, they would be expected to apply to only a small fraction of the data. In addition, since geo-

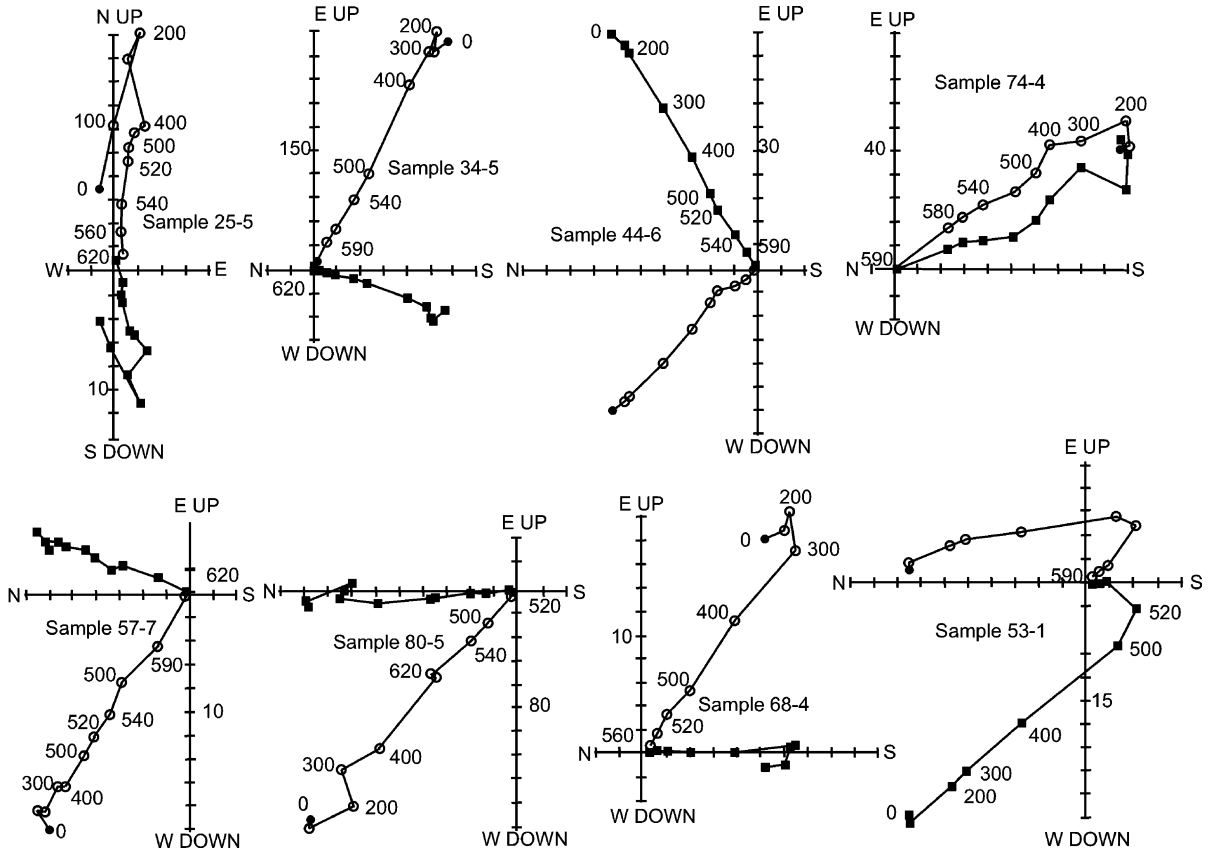


Fig. 5. Examples of typical thermal demagnetisation results from rocks of this study. Figures are temperatures in °C. Other symbols are as in Fig. 4.

magnetic field intensity drops to only ~ 10–20% of the typical dipolar field, rocks magnetised during a transition are likely to have weak NRMs and might be recognisable by this criterion. A tectonic interpretation appears to apply to all but three of the sites in the vicinity of Afyon city. Since these sites lie on a major fault system bordering the southern margin of the ESE–WNW-trending Afyon-Akşehir Graben (Fig. 2) where neotectonic deformation is continuing at the present day, we interpret these rotations as the cumulative effects of tectonic deformation bordering this graben since Miocene times.

Results from four Neogene volcanic fields (Fig. 2) are summarised in Sections 5.1–5.4 below. In addition, there are two outliers to consider plus a set of aberrant results in the vicinity of Afyon city (Sections 5.5–5.7):

5.1. SW Afyon

This is a field of basaltic lavas and alkaline derivatives extending for 50 km southwest of Afyon (Fig. 2). A range of age determinations between 14 and 8 Ma dates volcanic activity as middle to late Miocene. The actual duration of volcanism, however, was probably much more restricted because all the lavas with the exception of one site (16) are of reversed polarity and are therefore likely to belong to a single chron. Consistent age determinations of 8.0, 8.6, and 8.8 Ma would be compatible with eruption during the relatively long reversed Chron C4r at 8.26–8.70 Ma. Alternatively, but less securely, ages of 10.35 and 11.65 Ma could indicate that the older 11.5–11.9 Ma reversed chron is relevant to this volcanic episode if the younger ages are influenced by

Table 3
Site mean palaeomagnetic results from the volcanics of the Afyon region, Central Anatolia

Site number	N/n	R	k	α_{95}	D	I	Polarity
1	7/4	3.97	92.2	9.6	200.9	8.3	I
2	7/7	6.99	966.2	1.9	187.9	−48.4	R
3	8/8	7.96	177.9	4.2	37.8	−32.8	I
4	7/5	4.91	43.5	11.7	93.5	51.2	N
5	4/3	2.97	57.8	16.4	94.7	−55.5	R
6	5/5	4.98	201.5	5.4	158.5	−50.7	R
7	6/5	4.99	423.8	3.7	238.0	−45.6	R
8	6/6	5.93	76.8	7.7	207.0	−55.7	R
9	8/8	7.88	57.6	7.4	173.3	−47.8	R
10	6/6	5.98	305.0	3.8	177.8	−42.6	R
11	6/6	5.98	209.3	4.6	184.4	−41.9	R
12	5/4	3.99	243.2	5.9	89.2	74.4	N
13	6/6	5.90	49.3	9.6	174.9	−40.9	R
14	7/5	4.87	31.3	13.9	215.9	−36.3	R
15	5/5	4.99	452.9	3.6	193.1	−40.8	R
16	7/6	5.98	267.8	4.1	24.7	63.6	N
17	4/4	3.99	275.3	5.5	202.3	67.7	I
18	7/7	6.95	118.5	5.6	199.3	−45.5	R
19	5/5	4.95	76.7	8.8	229.3	−48.0	R
20	6/6	5.97	181.2	5.0	81.7	29.4	I
21	6/5	4.93	60.4	9.9	89.9	68.2	N
22	7/7	6.97	172.9	4.6	61.6	−23.1	I
23	6/6	5.91	58.4	8.8	74.5	7.8	I
24	5/4	3.81	15.7	24.0	288.6	3.7	I
25	7/7	6.97	183.5	4.5	172.5	−59.0	R
26	7/7	6.99	820.5	2.1	219.1	−22.2	R
27	7/7	6.89	56.4	8.1	204.2	−29.5	R
28	7/6	5.96	117.9	6.2	56.6	35.3	N
29	7/7	6.95	116.7	5.6	44.5	24.9	N
30	7/7	6.95	128.7	5.3	161.5	74.9	I
31	7/7	6.95	116.6	5.6	192.6	−36.8	R
32	7/7	6.94	92.9	6.3	178.1	−4.5	I
33	8/8	7.93	93.7	5.8	185.3	33.7	I
34	7/6	5.97	199.7	4.8	191.0	−56.6	R
35	7/7	6.94	95.3	6.2	197.0	−60.3	R
36	7/5	4.98	169.4	5.9	12.3	67.5	N
37	6/5	4.96	95.2	7.9	227.0	−45.7	R
38	7/7	6.98	357.3	3.2	207.9	−50.0	R
39	7/6	5.96	129.9	5.9	203.6	−51.6	R
43	7/7	6.96	135.4	5.2	57.6	41.4	N
44	7/7	6.93	87.3	6.5	62.2	22.5	N
45	7/5	4.98	240.6	4.9	14.2	40.2	N
47	8/8	7.99	596.5	2.3	209.0	−47.2	R
48	7/7	6.99	1034.2	1.9	5.9	35.0	N*
49	7/5	5.00	839.6	2.6	8.2	36.9	N
50	7/7	6.99	559.1	2.6	23.6	41.3	N*
51	7/7	6.99	485.5	2.7	34.7	47.1	N
52	7/7	6.93	82.8	6.7	24.5	7.7	N*
53	7/6	5.96	124.7	6.0	175.8	−29.8	R
54	7/6	5.55	11.2	20.9	152.9	13.5	I
55	7/6	5.98	314.4	3.8	143.0	−47.1	R
56	7/5	4.95	83.6	8.4	237.1	−54.3	R
57	7/7	6.96	141.9	5.1	14.0	40.6	N
58	7/7	6.98	364.2	3.2	10.6	40.4	N
59	7/7	6.95	112.1	5.7	29.2	40.8	N*
60	7/5	4.98	196.9	5.5	9.1	41.8	N*
61	7/7	6.99	539.6	2.6	7.8	52.8	N
62	7/7	6.98	346.9	3.2	353.4	57.1	N
63	7/6	5.97	176.8	5.1	11.1	63.3	N
64	7/5	4.98	210.6	5.3	31.3	38.7	N*
65	7/7	6.77	26.4	12.0	275.8	3.9	I
66	7/6	5.96	117.3	6.2	68.2	−32.7	I
67	7/5	4.93	53.8	10.5	244.9	−40.0	R
68	7/7	6.95	118.0	5.6	168.0	−48.7	R
69	5/5	4.99	405.3	3.8	191.0	−61.7	R
70	7/7	6.97	212.1	4.2	20.9	63.3	N
71	5/5	4.95	73.1	9.0	350.7	58.7	N
72	7/7	6.99	524.5	2.6	19.0	42.5	N
73	7/7	6.93	86.2	6.5	211.8	−23.4	R
74	6/5	4.96	112.7	7.2	160.2	−30.6	N
75	7/7	6.98	271.9	3.7	231.1	−53.7	R
76	7/7	7.00	1278.0	1.7	40.2	49.8	N
77	6/4	3.99	221.4	6.2	72.6	62.6	I
78	7/5	4.99	405.1	3.8	179.8	−63.4	R
79	7/7	6.99	485.8	2.7	188.8	−26.6	R
80	7/7	6.98	245.2	3.9	359.5	49.8	N
81	7/6	5.86	36.9	11.2	39.6	−53.9	I
82	7/7	6.92	75.1	7.0	61.7	0.5	I

D and *I* are the mean declination and inclination, respectively, derived from *n* samples out of a total site population of *N*. *R* is the magnitude of the resultant vector, and *k* is the Fisher precision parameter (=$(n-1)/(n-R)$); α_{95} is the radius in degrees of the cone of 95% confidence about the mean direction. Directions are classed as normal, reversed or intermediate (N, R or I) with respect to the present dipole axis.

* Tilt adjusted directions used in area and regional mean analysis (see text).

argon loss (Cande and Kent, 1995). The mean direction of magnetisation derived from 10 lavas ($D/I = 192/-49°$, $\alpha_{95} = 11°$) identifies small clockwise ro-

Table 4
Group mean (reversed polarity) palaeomagnetic results from volcanic rocks of the Afyon region

Group/volcanic field	N	R	α_{95}	k	D	I
SW Afyon	10	9.56	10.9	20.5	192.2	−48.8
Emirdağ	11	10.07	14.6	10.7	205.1	−42.1
İşcehisar	7	6.78	11.7	27.7	196.5	−46.5
İhsaniye	9	8.82	7.7	45.7	197.0	−44.0
İhsaniye area outliers	13	12.27	10.6	16.4	191.5	−55.7
Regional Mean	59	55.39	4.8	16.1	198.7	−47.5

For symbols, see Table 3.

Table 5
Neogene reference (reversed polarity) palaeomagnetic directions in Central Anatolia determined from mean apparent polar wander paths of adjoining major plates

Time (Ma)	Africa			Eurasia		
	D	*I*	α_{95}	*D*	*I*	α_{95}
0	180.0	−58.5	–	180.0	−58.5	–
10	185.5	−54.5	2.3	186.4	−55.2	2.2
20	186.4	−52.7	2.3	188.4	−54.4	3.3

tation since late Miocene times (Table 5). Sites 4, 5, and 17 yield aberrant directions but do not appear to identify transitional field directions because their intensities of magnetisation are comparable to the remainder of the collection (Table 1). They therefore probably record large local block rotations. Small outliers of this outcrop were sampled at basalt lava sites 73–75 (Fig. 1). All are of reversed magnetisation and 73 and 75 are rotated clockwise. However, since sites 73 and 74 have anomalously shallow inclinations and both lie close to a fault zone, it seems likely that they too, have been subject to local tectonic rotation; in this case, correction for dip of lava flow structure brings the direction close to the predicted inclination ($D/I = 218/-51°$).

5.2. Emirdağ

There is no age control for this andesite volcanic field on the northeastern margin of the Afyon-Akşehir Graben and at the eastern limit of the present survey (Fig. 2), but mixed polarities are present (8R, 3N) and imply that activity spanned more than one chron. Intermediate directions at sites 24, 30, and 33 are no more weakly magnetised than the remaining sites

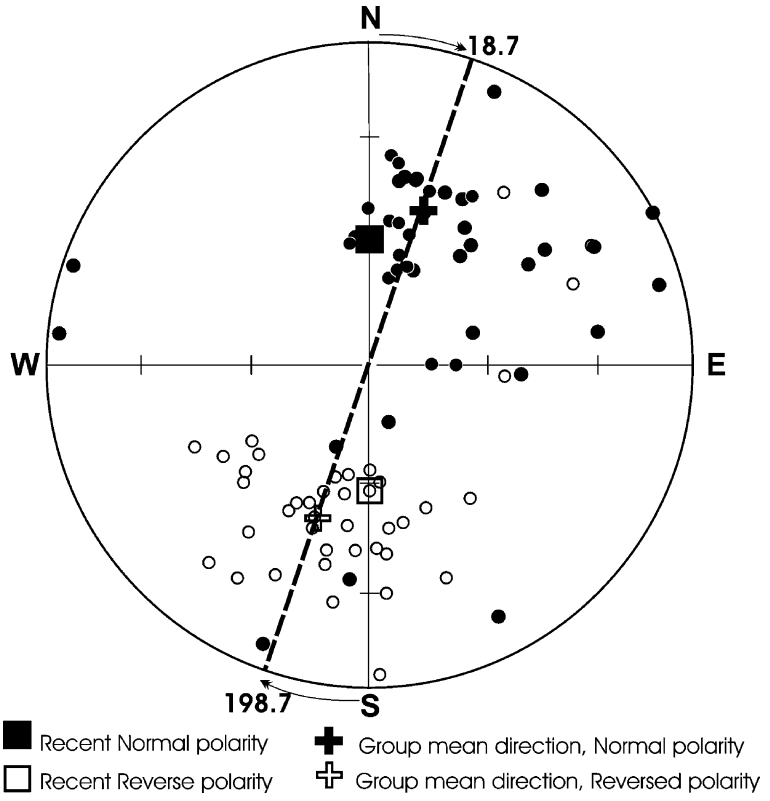


Fig. 6. Site mean directions of magnetisation from volcanic units of this study. Open symbols are upper hemisphere plots, and closed symbols are lower hemisphere plots. The squares are the directions of the present-day dipole field in this region, and the cross is the overall mean palaeomagnetic field direction derived from the Afyon region.

(Table 1) and therefore probably record local block rotations. The mean reversed field direction derived from 11 units is $D/I = 205/-42°$ ($\alpha_{95} = 15°$), which is not significantly different from the mean direction of the SW Afyon field on the opposite shoulder of the graben.

5.3. İşcehisar

Small volcanic outcrops in this region are both basaltic flows and ignimbrite sheets. They are cut by WNW–ESE normal faults parallel to the trend of the Afyon-Akşehir Graben, but all are clearly of normal or reversed magnetisation (Table 2). The mixed polarities observed here show that activity spanned at least one polarity inversion and the single age determination of 11.6 ± 0.2 Ma would be consistent with frequent reversals between chrons C5n.1n and C5An.1n (Cande and Kent, 1995). The mean reversed direction for the volcanic field is $D/I = 197/-47°$ ($\alpha = 12°$). Two more strongly rotated sites (43 and 44) lie close to a fault plane (Fig. 2).

5.4. İhsaniye

The extensive volcanic field east of this town comprises ignimbrites and acid tuffs. It evidently represents a short igneous episode because all of the sites are of normal polarity; none are transitional or subject to significant local rotation. The age determinations near the northern rim of the outcrop suggest that one of the relatively long chrons C5Dn at 17.28–17.62 Ma or C6n at 19.0–20.1 Ma is the interval recorded here. This is probably the earliest Neogene volcanism in the Afyon region, although the mean remanence direction ($D/I = 197.0/-44.0°$, $\alpha_{95} = 8°$) identifies a similar rotation to the volcanic fields to the SE and NW.

5.5. Outliers to the north of the İhsaniye field

These comprise distributed small fields of andesite and basaltic lavas and are probably appreciably younger than the acid volcanics to the south (Fig. 2). Units of both normal and reversed magnetisation are present (Table 2) with two sites with intermediate directions (65 and 66) subject to local rotation in the easternmost outcrop near Seyitgazi. The area mean derived from all of these outcrops is $D/I = 192/-56°$ ($\alpha = 11°$) and shows comparable clockwise rotation to the fields further to the south, although the inclination is significantly steeper and close to the expected mean direction of the geomagnetic field in this region.

5.6. The vicinity of Afyon City

These volcanics (sites 18–25, 77, 80–82) are distinguished from the remainder of the collection because all but 3 of the 12 sites yield anomalous directions of magnetisation. The rock units are massive andesites in which clear indicators of the palaeohorizontal are absent (Table 1), and they are sited at a complex offset pattern in the major fault bounding the southwest margin of the Afyon–Akşehir Graben (Fig. 2). The magnetisations have relatively high intensities, and there is no indication that they record transitional field directions. They are therefore attributed to large tectonic rotations along this WNW–ESE-bounding fault system (Fig. 2). Most inferred rotations are strongly clockwise (60–90° at units 12, 20, 21, 23, 77, and 82) and would be compatible with right-lateral transtension across the graben, but there has also been shallowing or, less commonly, steepening of the remanence direction, indicating that back rotations have also occurred about inclined axes (Table 2).

5.7. Miscellaneous outcrops

Two sites (1 and 2) sampled in ignimbrite sheets at Hoyran Lake at the southern limit of the study area (see Koçyiğit, 1984) are of reversed magnetisation and also rotated clockwise. Site 1 lies close to a fault zone and has apparently been rotated about an inclined axis.

6. Regional rotation deduced from palaeomagnetic study

Tectonic rotation, R', in deformed terranes is determined by comparison of the mean directions with the predicted palaeofield axis from an adjoining stable

terrane. The study shows that rotation across this region is consistently clockwise by ~10° from the mid-Miocene field. Both observed and reference directions have confidence limits (ΔD and ΔD_{ref}, respectively). Beck (1980) proposed a confidence limit on R' defined by $\Delta R' = \sqrt{(\Delta D^2 + \Delta D_{ref}^2)}$. Demarest (1983) concluded that this value overestimates errors and showed that provided α_{95} is small and preferably less than 10°, a standard correction factor is applicable. For $n \geq 6$, the correction factor lies between 0.78 and 0.80; $\Delta R'$ can then be derived from the equation $\Delta R' = 0.8\sqrt{(\Delta D^2 + \Delta D_{ref}^2)}$. Only if $R' > \Delta R'$ can a rotation be regarded as significant. Assuming an average age of 10 Ma and comparing the regional mean direction from the volcanic fields of the Afyon district (Fig. 6) with the local direction derived from the Eurasian APWP (Besse and Courtillot, 1991) yields a cumulative clockwise rotation of $12.3 \pm 4.2°$ in the Afyon region with respect to Eurasia and $13.2 \pm 4.2°$ with respect to Africa. Intense local tectonic rotations are also recognised in the study region, but are largely confined to the margin of the Afyon–Akşehir Graben. This latter finding is comparable to the outcome of palaeomagnetic studies of crustal deformation elsewhere in Turkey, where large Neogene rotations implying rates in excess of 10°/100,000 years have been detected but are confined to small tectonic flakes a few kilometres in width between major faults defining the intracontinental transforms (Tatar et al., 1995; Piper et al., 1997; Gürsoy et al., 2003).

The rotations are comparable in rock units ranging from ~18 to ~8 Ma in age. Hence, regional rotation occurred after all the recorded volcanism

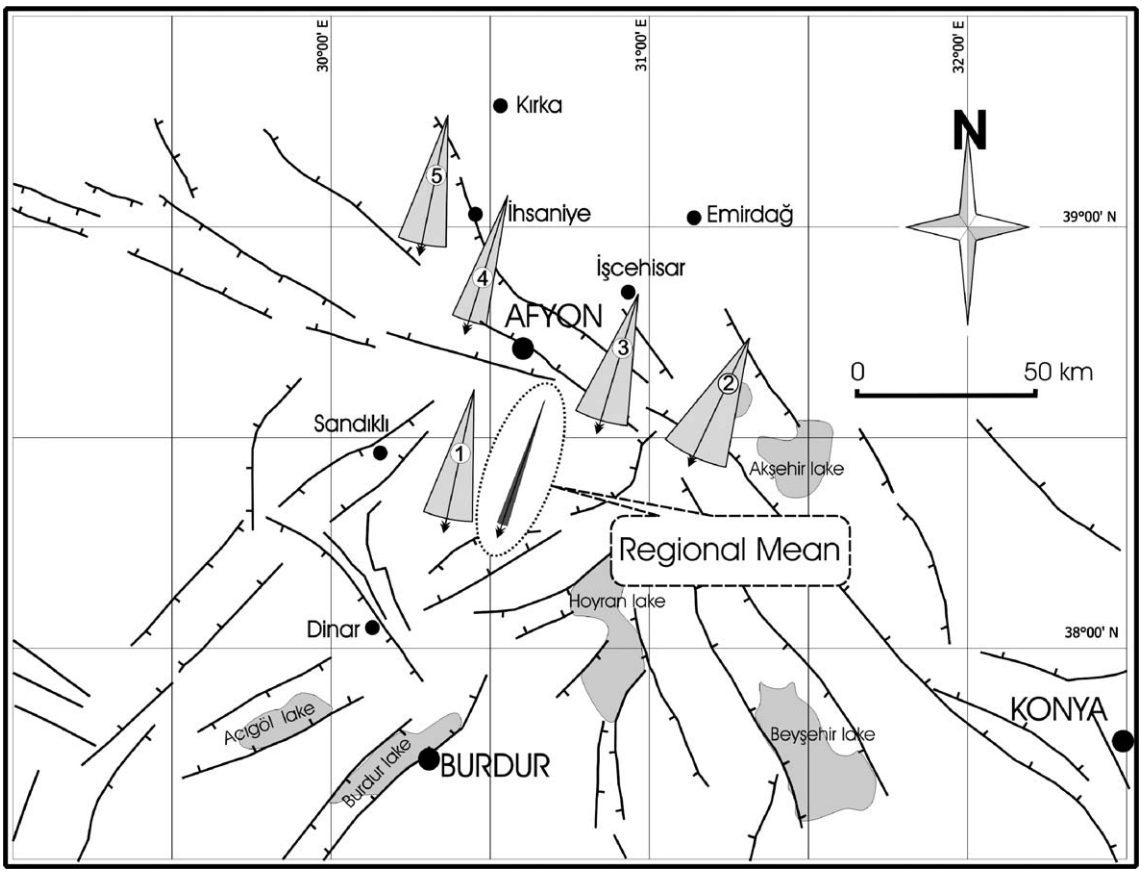


Fig. 7. Regional distribution of group mean directions of magnetisation in the Afyon region with 95% confidence limits.

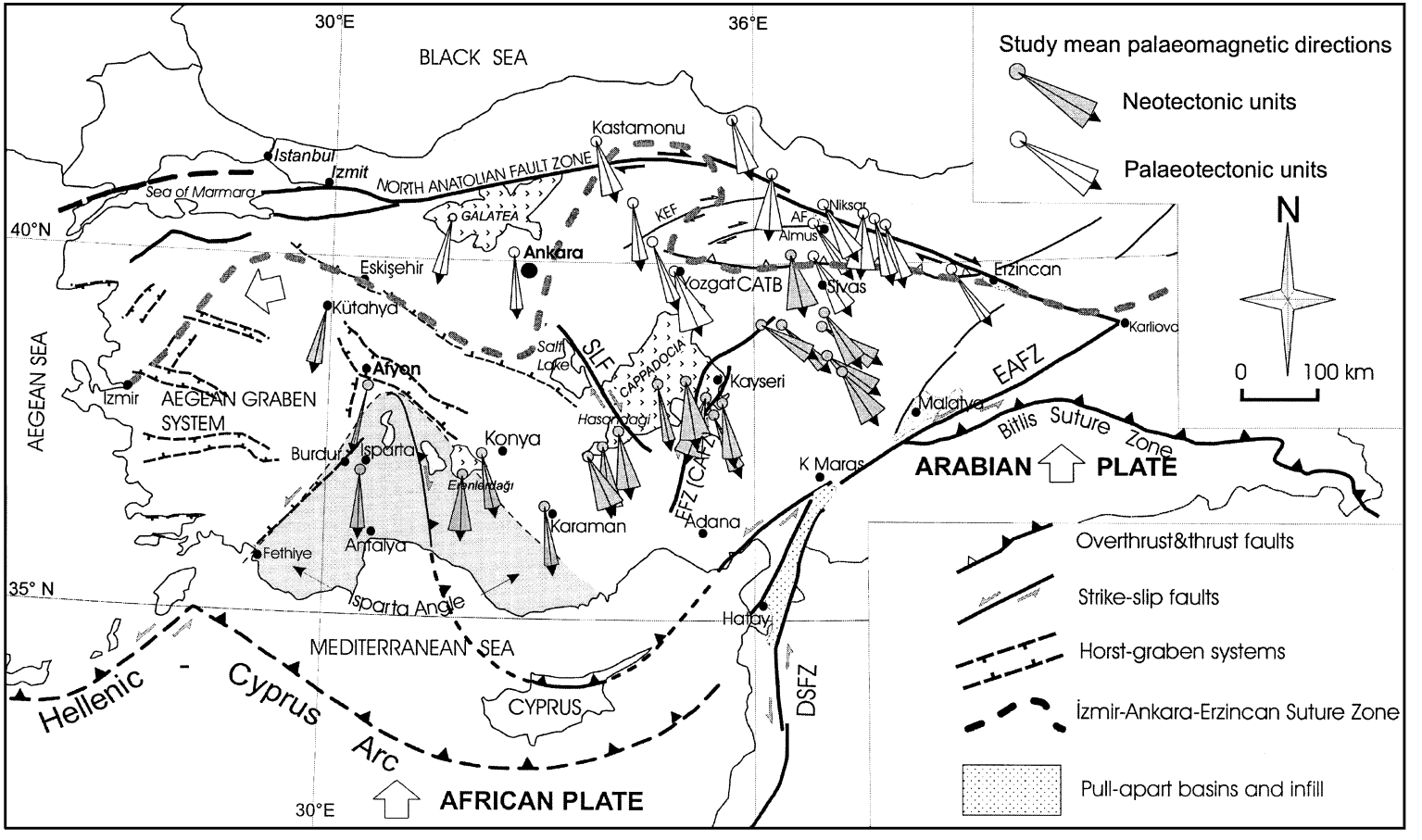


Fig. 8. Schematic tectonic map of Turkey showing distribution of major faults active during the Neotectonic regime and declinations of mean palaeomagnetic vectors with cones of 95% confidence from Late Cretaceous and Cenozoic rock units adjusted for tectonic tilts (see Table 6). All directions are shown with reversed polarity. The thick dashed (IAEZ Zone) line is the site of Neotethyan convergence during the palaeotectonic era.

Table 6
Summary of Cenozoic palaeomagnetic results from Central Turkey

District/Fault block	Age	Age (Ma)	Location		N	D/I	α_{95}	Reference
			E	N				
(a) Results from rock units emplaced during the palaeotectonic regime								
North of the NAFZ								
Mesudiye (N)	E		37.7	40.6	7	165.9/−50.4	6.7	Baydemir (1990)
Mesudiye (R)	E		37.7	40.6	8	185.6/−45.1	15.1	Baydemir (1990)
Niksar	E	41	37.0	40.7	10	152.4/−42.5	11.3	Tatar et al. (1995)
Erbaa	E		36.3	40.8	7	194.6/−48.8	15.3	Piper et al. (1996)
Mesudiye	E		37.7	40.5	7	166.8/−49.9	9.6	Orbay and Bayburdi (1979)
Kusuri	C–E		36.3	41.4	5	161.2/−48.4	12.9	Sarıbudak (1989)*
Kastamonu	E?		33.7	41.2	3	164.2/−31.7	10.6	Piper et al. (1996)
South of the NAFZ								
Imranli	E		38.4	39.8	10	146.0/−34.2	6.3	Baydemir (1990)
Almus	E		36.9	40.4	8	144.1/−47.5	7.6	Tatar et al. (1995)
Kalehisar	E–M		34.5	40.2	5	160.4/−51.0	5.2	Piper et al. (1996)
Akdagmadeni	E		35.8	39.7	3	133.9/−43.5	5.2	Tatar et al. (1996)
Yozgat	E		34.7	39.8	5	158.4/−46.5	18.6	Tatar et al. (1996)
Bayat	E		34.5	40.8	7	172.1/−47.1	7.6	Piper et al. (1996)
Galatea	M		31.8	40.6	15	196.8/−56.5	6.9	Gürsoy et al. (1999)
(b) Results from rock units emplaced during the neotectonic regime								
CAT	M–Q		36.5	39.8	5	169.7/−47.6	19.9	Gürsoy et al. (1997)
Yildizeli	M–Q		36.9	39.8	12	150.9/−50.8	11.9	Gürsoy et al. (1997)
Şarkışla	M–Q	13.8–13.1	37.0	39.5	18	147.9/−55.0	10.9	Gürsoy et al. (1997)
Kangal	M–Q		37.0	39.0	6	129.6/−47.6	15.6	Gürsoy et al. (1997)
Gürün	P–Q		37.2	38.8	7	157.0/−54.1	14.0	Gürsoy et al. (1997)
Gemerek, R	M	15.2–14.8	36.0	39.2	35	111.9/−60.8	9.6	Krijgsman et al. (1996)
Gemerek, N	M	15.2–14.8	36.0	39.2	41	127.9/−32.2	6.7	Krijgsman et al. (1996)
Inkonak, R	M	25.8–25.1	37.0	39.4	30	129.8/−37.3	7.7	Krijgsman et al. (1996)
Inkonak, N	M	25.8–25.1	37.0	39.4	30	127.1/−49.6	10.0	Krijgsman et al. (1996)
Yeniköy, R	M	30.4–28.4	36.4	39.1	49	126.3/−33.3	8.8	Krijgsman et al. (1996)
Yeniköy, N	M	30.4–28.4	36.4	39.1	32	149.6/−47.7	7.2	Krijgsman et al. (1996)
Haramiköy, R	M	23.0–21.8	31.8	38.5	37	190.1/−40.9	5.3	Krijgsman et al. (1996)
Afyon	M	12.3–8.8	30.7	38.8	12	198.7/−47.5	9.7	this study
Sille Volcanics	M	11.6	32.1	37.7	4	149.5/−51.8	11.3	Tatar et al. (2002)
Erenlerdağ (i)	M	9.9	32.1	37.7	10	182.5/−47.0	9.7	Tatar et al. (2002)
Kütahya (i)	M		29.9	39.6	5	205.4/−54.6	13.0	authors, in preparation
Kütahya (ii)	M		29.9	39.6	6	205.3/−39.8	11.8	authors, in preparation
Erenlerdağ (ii)	M–P	6.9–3.5	32.0	37.6	5	179.0/−50.8	13.4	Tatar et al. (2002)
Hasandağ	M	13.1	34.2	38.0	7	170.1/−57.6	10.9	Gürsoy et al. (1998)
Ignimbrites	M–P	11–1.1	34.8	38.6	10	174.0/−51.1	6.6	Piper et al. (2002)
Isparta	P	4.7–4.0	30.8	37.8	15	185.8/−52.6	6.1	Tatar et al. (2002)
Karaman	M–P	6.0–4.7	33.2	37.2	13	174.3/−51.8	8.6	Gürsoy et al. (1998)
Kütahya (iii)	P		29.9	39.6	9	205.4/−53.1	6.1	authors, in preparation
Kütahya (iv)	P		29.9	39.6	6	161.6/−41.5	9.1	authors, in preparation
Karacadağ	P	1.5	33.7	37.7	13	177.5/−57.6	7.0	Gürsoy et al. (1998)
Karapınar	Q		33.6	37.5	5	156.9/−42.2	14.9	Gürsoy et al. (1998)
E. Erciyes	Q	<1.1–0	35.6	38.6	7	173.1/−52.5	9.4	Tatar et al. (2000)
W. Erciyes	Q	<1.1–0	35.0	38.5	8	170.1/−49.7	13.2	Tatar et al. (2000)
Cappadocia	Q	<1.1–0	34.7	38.3	6	189.7/−53.2	16.0	Tatar et al. (2000)
Sultansazlığı (i)	Q	<1.1–0	35.2	38.5	5	202.2/−52.2	16.0	Tatar et al. (2000)
Sultansazlığı (ii)	Q	<1.1–0	35.5	38.4	5	159.9/−56.4	9.4	Tatar et al. (2000)

and is therefore a late feature of the neotectonic evolution (Fig. 7). Whilst the palaeomagnetic data cannot by themselves discriminate between the four tectonic models discussed in Section 2, they clearly favour an episodic framework with rotation concentrated during the last phase. Due to the absence of younger rocks, it does not appear to be possible to constrain rotation within the last 8 Ma in this region, although in Cappadocia and the Sivas Basin further to the east, block rotations linked to tectonic escape are concentrated within the last 3 Ma, and probably within the last 1.5 Ma (Gürsoy et al., 1997; Piper et al., 2002). Although Platzman et al. (1998) and Kissel et al. (2003) conclude that rotation in these regions has been progressive since Miocene times, their conclusion is based on small numbers of scattered sites and must be discounted in the light of the much more detailed studies summarised in Fig. 8 and Table 6.

The magnetic inclinations are mostly $\sim 10^\circ$ shallower than the regional field predicted from Eurasia during the Miocene (Tables 4 and 5). However, the studied volcanics were all emplaced after the palaeotectonic phase of Neotethyan closure and continental collision, and this inclination anomaly cannot be attributed to the ~ 1000 km of relative northward motion since the Miocene (Piper et al., 2002). Only a small fraction of this motion would be required to accommodate crustal thickening in Anatolia, whilst the evolution of the extensional province to the west would suggest that motions here have in fact been southerly and behind a retreating Hellenic Trench (Angelier et al., 1982). A geomagnetic field source more complex than a geocentric dipole has been invoked to explain the time-averaged properties of the geomagnetic field in Late Tertiary times (Wilson, 1971). However, an offset dipole or multipolar field source cannot be a satisfactory explanation for the inclination anomaly recognised here (and elsewhere in the eastern Mediterranean) because complexities in the field source would also be reflected in the data used to calculate the reference paths of the adjoining major plates (Beck et al., 2001). Hence, a geomagnetic anomaly of regional rather than global extent is probably required to explain the inclination anomaly.

7. The palaeomagnetic signature of tectonic escape

The present record of Cenozoic palaeomagnetic results from Anatolia is summarised in Table 6 and is illustrated in Fig. 8. This summary includes results from units emplaced during the palaeotectonic regime of terrane accretion that may therefore incorporate a component of rotation prior to the neotectonic regimes of strike–slip and extension. Most results from these older units, however, yield declinations comparable with results from late Miocene and younger units and suggest that most, or all, of the declination anomaly is attributable to rotation during the neotectonic regime. In central and northern Anatolia, magnetic declinations in Eocene units are rotated anticlockwise on both sides of the NAFZ, and a component of regional rotation in this area must therefore predate the initiation of this fault in late Pliocene times. However, in the Almus region, Tatar et al. (1995) identify 26° of anticlockwise rotation in Eocene rocks on the north side of the fault zone and 34° on the south side. Further to the east, Baydemir (1990) also used rocks of Eocene age to demonstrate anticlockwise rotations, in this case, 44° on the north side and 53° on the south side of the fault zone. Both studies indicate that long-term differential rotation is occurring by extrusion of blocks along lateral side splays to the NAFZ master fault. Whilst the bulk of these rotations in northern Anatolia occurred before initiation of the NAFZ, further to the south in the Sivas Basin, comparable amounts of rotation are found in rocks as young as Quaternary (Gürsoy et al., 1997) and were therefore confined to the last few million years. In the Cappadocian ignimbrite succession, a uniform low rate of

Notes to Table 6:

Directions given are reversed palaeofield directions and locations are the centre of the study area. N is the number of separate units (sites = flow units or dykes) included in the calculated mean, and α_{95} is the radius of the cone of 95% confidence about the mean direction. CAT = Central Anatolian Thrust Belt.

* Mean calculated for five sites in Late Cretacous–Eocene volcanics. Note that additional results from distributed isolated sites across this region are given by Platzman et al. (1994, 1998) and Kissel et al. (2003).

anticlockwise rotation (~ 1°/Ma) occurred between 8.2 and 1.1 Ma and subsequently accelerated following cessation of volcanism in this region (Piper et al., 2002). The cumulative rotation diminishes progressively to the west from the Sivas Basin (Gürsoy et al., 1997) to Cappadocia (Piper et al., 2001, 2002), to the Karaman and Hasandağı regions (Gürsoy et al., 1998), and finally to the Erenlerdağ region where no significant rotation is evident (Tatar et al., 2002).

The regional distribution of anticlockwise rotation shown in Fig. 8 shows that the largest rotations are concentrated near the pincer zone where the northern perimeter of the Arabian Plate has impinged into the Anatolian collage. Rotation diminishes rapidly in the broader lower strain region of accreted Anatolian terranes to the west. The major tectonic boundaries at the southeastern margin of the Anatolian collage are the subparallel Ecemiş and East Anatolian Fault (EAF) zones (Fig. 8). The former exhibits lower levels of seismic activity than the latter (Demirtaş and Yılmaz, 1996), but is interpreted to be a sinistral intracontinental transform by Kocyiğit and Beyhan (1998) and designated the Central Anatolian Fault Zone (CAFZ). With the present distribution of studies, it is not clear whether substantially differing amounts of tectonic escape have occurred on either side of the CAFZ, although Jaffey and Robertson (2001) found that Plio-Quaternary motion across the CAFZ is mainly extensional. They speculate that the anticlockwise rotation in this sector has rotated the fault zone away from an optimal orientation for accommodating strike–slip motions during tectonic escape. Significant differential neotectonic rotations are not evident across the northeastern continuation of this fault zone (Fig. 8) and suggest it has not been a site of major differential motion. Cumulative rotations on the west side of the CAFZ from Cappadocia to Erenlerdağı are lower in magnitude and broadly follow the southern protrusion of the Tauride Arc. The radial pattern of neotectonic declinations across this region from anticlockwise in Cappadocia and Karaman to near zero in Erenlerdağı and then clockwise into the Isparta Angle (Tatar et al., 2002) and the Afyon region (this paper) is consistent with an expansion in the curvature of this arc during Quaternary times.

The regional rotation determined in the Afyon region is larger than the 4° of rotation deduced from the Plio-Quaternary Isparta volcanic field to the south (Tatar et al., 2002) and slightly larger than the 10° of post-Miocene rotation deduced from the Galatean Volcanic Province to the NNE (Fig. 8 and Gürsoy et al., 1999). This would be consistent with an expansion of the extruded terrane perimeter to the west as well as to the south, and it further implies that tectonic escape contributes to producing the extensional regime in the west. However, the present velocity of the crust in the Afyon region deduced from GPS is ~ 2 cm/year to the WSW relative to Eurasia, and a comparable motion over a wider area of central implies anticlockwise motion about a remote pole of rotation in the SE Mediterranean region (Reilinger and McClusky, 2001). This analysis of contemporary motions from GPS clearly contrasts with the long-term anticlockwise rotation about a local axis deduced from palaeomagnetism. The rate of rotation deduced from palaeomagnetism is also significantly larger than the mean rate of contemporary rotation of 1.3 ± 0.1°/Ma deduced from GPS. This discrepancy between the short-term assessment derived from GPS and the long-term assessment deduced from palaeomagnetism has several possible explanations. It could reflect long-term temporal changes in the regional stress field such as those highlighted in Section 2, or it could arise because the GPS results do not incorporate anelastic deformation or embrace sufficiently long periods to be representative of geological time.

Although the regional coverage of neotectonic rotations is still incomplete (and is indeed largely constrained by the distribution of young volcanics), the picture that is emerging is one of distributed crustal deformation with considerable regional variation (Fig. 8). Most rotation probably occurred in the south after the north because in the latter region, the bulk predates initiation of the NAFZ. In the south, most rotation has occurred during the last 1–2 Ma and presumably following prolonged crustal thickening following late Miocene collision with Arabia along the Bitlis Zone. The magnitude of rotation diminishes dramatically to the west away from the Arabian pincer and suggests that tectonic escape is being dispersed between the Galatean region bordering the NAFZ in the north (Gürsoy et al., 1999) and the Tauride Arc in the south. Rotations appear to involve blocks on a 10–100-km scale (Fig. 8), with the higher part of this range implied in the Afyon

area, and there is no substantial indication that the Anatolian region has acted as a coherent plate on a time scale of millions of years. The view that such a plate has undergone discrete episodes of rotation (Platzman et al., 1998; Kissel et al., 2001) is unsupported by the wider regional evidence summarised in Table 6 and Fig. 8.

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